World Population Trend and CO_2 Emission

NIYIGENA Fiston 2025MBI041

November 2026

Abstract

This project examines population trends and their relationship with carbon dioxide (CO_2) emissions using the supplied World Population and CO_2 Emissions datasets. The work involved data inspection, missing-value and duplicate checks, outlier identification, population projection to 2030, visualisation, correlation, regression and comparison by continent. The population dataset contains 234 countries and territories and 17 variables; the CO_2 dataset contains 215 country records and annual per-capita observations from 1990 to 2019. The population projection applies the exponential model and assumes a constant annual growth rate. After matching country names and excluding missing 2019 CO_2 values, 168 countries remained. The Pearson correlation between 2022 population and 2019 CO_2 emissions per capita was 0.0082 ($p = 0.9163$), indicating virtually no linear association in this sample. Asia had the largest sum of country-level per-capita values in the continent comparison. These sums are not total continental emissions. The findings show that population size alone does not explain differences in CO_2 emissions per person.

1. Introduction

Population growth can increase demand for energy, transport, goods and other resources, which may contribute to CO_2 emissions. However, emissions also depend on energy sources, economic and industrial structure, technology and consumption patterns. This project investigates population trends and assesses whether countries with larger populations also have higher CO_2 emissions per person. It follows the examination tasks on exploratory analysis, population projection, visualisation and correlation analysis.

2. Methods and dataset

The World Population dataset contains 234 rows and 17 columns, including country/territory, continent, population at selected years (1970–2022), area, density, growth rate and world population percentage. The CO_2 dataset contains 215 country records, with annual values from 1990 to 2019 for the indicator "CO_2 emissions (metric tons per capita)." The duplicate 2019 field in the supplied CO_2 file was excluded, retaining the first 2019 column. The datasets were joined using country names.

Data preparation included inspecting dimensions, variable names, sample records and data types; checking duplicate records and missing values; and using separate boxplots and the 1.5×IQR rule to identify potential outliers. The population data had no missing values or duplicate rows. Missing CO_2 observations were retained as missing rather than replaced with zero. Outliers were flagged but not automatically removed because large differences in country population and area can be genuine.

For the 2030 projection, the model was

$$N_t = Pe^{rt},$$

where P is 2022 population, $t = 8$ years and r is the annual rate. Since the supplied growth-rate values are factors around 1, the annual rate was interpreted as `Growth Rate`-1. This assumes a constant rate through 2030. The analysis used graphs for the ten most populous countries, Pearson correlations, a linear regression and a continent-level bar chart.

3. Results and findings

3.1 Exploratory analysis and outliers

The population file contained 234 rows and 17 columns, with no missing values or duplicate records in the supplied copy. The CO_2 file contained 215 country records and some missing country-year observations. The IQR method identified potential outliers in several numeric population variables: 25 in 2022 population, 26 in 2020, 27 in 2015, 26 in 2010, 29 in 2000, 31 in 1990, 32 in 1980, 29 in 1970, 28 in area, 22 in density and 3 in growth rate. Rank had no outliers. These observations were retained because they may represent real differences between countries.

3.2 2030 population projection

The largest projected populations were India (about 1.496 billion), China (1.426 billion), the United States (348.7 million), Indonesia (290.0 million), Pakistan (274.8 million), Nigeria (265.0 million), Brazil (223.4 million), Bangladesh (186.6 million), Ethiopia (151.5 million) and Russia (141.6 million). The estimates are mathematical projections under a constant-growth assumption, not official forecasts.

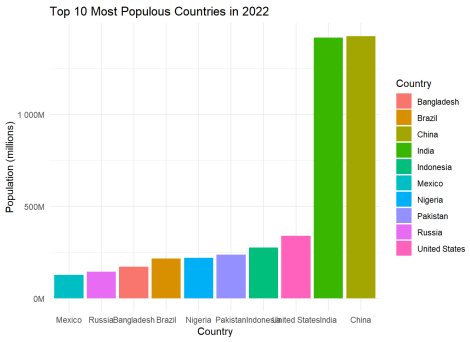


Figure 1: Top 10 countries by 2022 population.

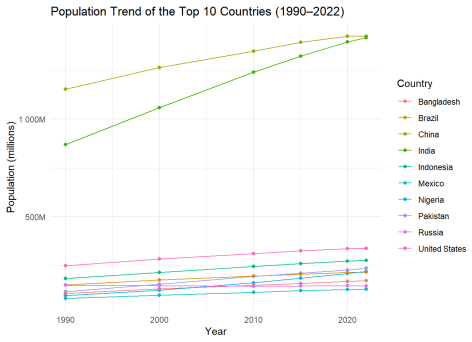


Figure 2: Population trends, 1990–2022.

3.3 Population and CO_2 trends

The ten largest populations in 2022 were China, India, the United States, Indonesia, Pakistan, Nigeria, Brazil, Bangladesh, Russia and Mexico. The population line graph compared the available benchmark years 1990, 2000, 2010, 2015, 2020 and 2022. The CO_2 line graph displayed available per-capita observations for these countries from 1990 to 2019. Because the CO_2 indicator is measured per person, it describes emissions intensity at country level rather than total national emissions.

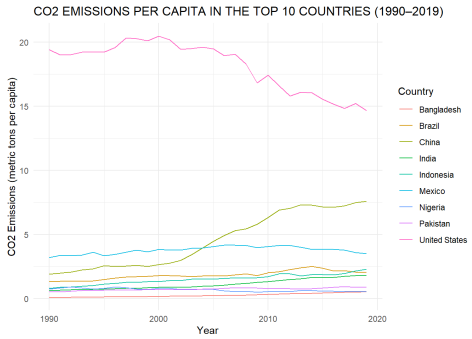


Figure 3: CO_2 emissions per capita for the top 10 countries, 1990–2019.

3.4 Correlation and regression

The population-variable correlation matrix showed a moderate positive correlation between area and world population percentage ($r = 0.45$). Other reported relationships were weak: area with density (-0.06) and growth rate (-0.01); density with growth rate (-0.07) and world population percentage (-0.03); and growth rate with world population percentage (-0.02).

After joining by country name and excluding missing 2019 CO_2 values, 168 countries were analysed. The correlation between 2022 population and 2019 CO_2 emissions per capita was $r = 0.0082$ ($p = 0.9163$; 95% confidence interval: -0.1434 to 0.1594). The regression also showed no meaningful linear relationship ($R^2 = 0.000067$, slope $p = 0.916$). Thus, population size was not significantly associated with per-capita CO_2 emissions in this sample.

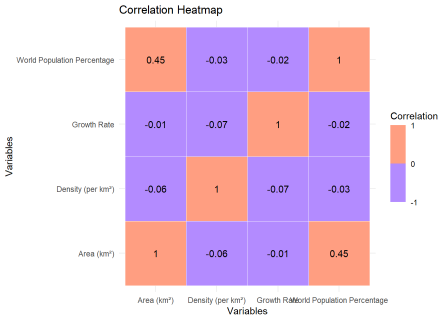


Figure 4: Correlation matrix.

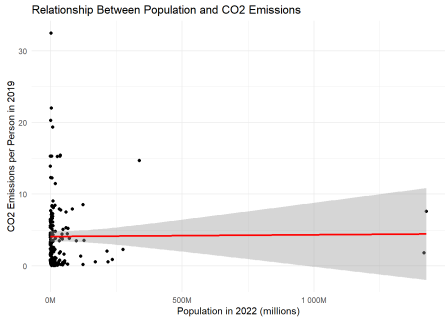


Figure 5: Population and CO_2 per-capita relationship.

3.5 Comparison by continent

The assignment's requested sums of country-level 2019 per-capita values were: Asia 242.03 (37 countries), Europe 229.18 (40), North America 79.57 (19), Africa 58.01 (48), Oceania 53.20 (13) and South America 29.42 (11). Asia ranked first, North America third, and South America last. These are sums of per-capita values, not total emissions by continent; adding per-capita rates does not produce a physically meaningful emissions total.

4. Discussion

The near-zero correlation and very low regression R^2 indicate that population size alone does not explain cross-country differences in CO_2 emissions per person. Countries with similar populations may rely on different energy sources and

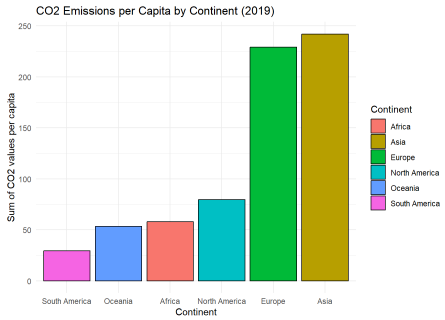


Figure 6: Continent comparison of summed country-level per-capita values, 2019.

have different industrial structures, income levels and transport systems. The results should be interpreted cautiously because some CO_2 observations are missing, matching depends on country names, and the population projection assumes constant growth. The continent comparison is limited by the use of summed per-capita values.

5. Lessons learned and recommendations

The project provided practical experience in data inspection, cleaning, missing-value assessment, outlier detection, reshaping, visualisation, dataset merging, correlation and regression in R. Future work should use ISO3 country codes for more reliable matching, obtain complete and consistent annual population data, and use total CO_2 emissions when examining the relationship between population size and total emissions. For comparisons of emissions per person across continents, population-weighted averages should be used instead of sums of country-level per-capita rates.

6. Conclusion

The supplied datasets enabled an exploration of population trends, 2030 population estimates and country-level CO_2 emissions per person. Although the population projection identifies countries expected to have the largest populations under the stated model, the statistical analysis found virtually no linear relationship between 2022 population size and 2019 per-capita CO_2 emissions. Population is only one factor among several that influence emissions. The findings must also be considered in light of missing CO_2 data, country-name matching and the limitations of the continent-level per-capita sums.